Y E L D E N   P R O T O C O L

YAAF

Yelden Agent Accountability Framework

A Mathematical Theory of Accountability for Autonomous Trading Agents

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Kelly Criterion · Cramér-Lundberg · Deflated Sharpe Ratio · CVaR · Systemic Risk

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Foreword

Autonomous agents are already managing capital. They execute trades, enter agreements, and allocate resources on behalf of humans — often faster than any human could supervise. The infrastructure to hold them accountable has not kept pace.

This e-book presents YAAF — the Yelden Agent Accountability Framework — a rigorous mathematical system that answers three questions no existing protocol addresses with reference to actual trading performance:

- How much must an agent stake to access vault capital?
- How much capital may it deploy given its track record?
- Under what conditions is stake destroyed?

YAAF is not a theoretical proposal. It is a deployed framework, live on Polygon mainnet since March 2026, validated against real trading data from an autonomous agent with 122 live trades, bootstrap Monte Carlo simulation across 5,000 paths, and four rounds of peer review.

The framework draws on foundational literature in mathematical finance, portfolio theory, and risk management: Kelly (1956), Cramér (1930), López de Prado (2014, 2018), Markowitz (1952), Rockafellar & Uryasev (2000), and the post-2008 systemic risk literature (Adrian & Brunnermeier, 2016). Each design decision is grounded in established theory and validated against empirical data.

> *YAAF is the first framework to derive capital allocation for autonomous AI trading agents from demonstrated edge — not fixed rules, not arbitrary limits, not reputation scores. The math decides.*

Chapter 1 — The Unaccountable Machine

1.1 The Autonomous Agent Economy

The deployment of autonomous AI agents in financial markets has accelerated dramatically since 2024. Agents built on large language models, reinforcement learning, and classical quantitative methods now execute trades, manage portfolios, and interact with DeFi protocols with minimal human oversight.

By early 2026, tokenized real-world assets (RWA) reached $30 billion in total value locked, with autonomous agents representing a growing share of active vault managers. The technology has outrun the accountability infrastructure.

> *"You can't lend to an agent you can't evaluate." — AgentProof, 2026*

1.2 Three Questions Without Answers

Existing accountability frameworks — Ethereum Proof-of-Stake (Buterin & Griffith, 2017), Cosmos slashing, EigenLayer service agreements — were designed for consensus participation, not trading performance. They answer none of the questions that matter for capital allocation:

Question	Existing Frameworks	YAAF
How much must an agent stake?	Fixed (32 ETH, arbitrary)	Kelly × CF × MDD_factor — from performance
How much capital may it deploy?	Fixed or variable, not performance-based	min(C_kelly, C_stage) — mathematically derived
When is stake destroyed?	Consensus violation, service SLA	DD + Capital Loss dual trigger — trading P&L
Anti-overfitting gate?	None	Deflated Sharpe Ratio (López de Prado, 2014)
Tail risk measurement?	None	CVaR + Skewness + Kurtosis penalties
Anti-gamification?	None	Volume Axiom — min position size enforced

1.3 The 24% Loophole

A critical failure mode in existing slash mechanisms: an agent reaches 24% drawdown, stops trading, and avoids the 25% slash threshold. Stake is preserved. The vault has already absorbed significant capital loss.

This is not an edge case — it is a rational strategy for any agent optimizing for stake preservation rather than vault performance. YAAF v1.2 closes this loophole with a dual-trigger slash function. YAAF v1.3 extends this to systemic scenarios.

1.4 The Liquidity Singularity

A second failure mode emerges at scale: when hundreds of autonomous agents share training data, optimization targets, or signal sources, their strategies converge. The simultaneous entry and exit of correlated agents creates what we term the Liquidity Singularity — a self-reinforcing collapse of market liquidity driven by algorithmic convergence.

Three cascade mechanisms:

- Fragmentation Illusion — a vault deploying across 100 correlated agents believes it has diversified but holds a single concentrated position fragmented across 100 execution nodes.
- Liquidity War — simultaneous exits compete for the same liquidity; agents with superior infrastructure extract value from slower agents; retail absorbs the residual cost.
- Slash Cascade — systemic drawdowns trigger simultaneous slash events; forced stake liquidation depresses the protocol token; remaining agents face deepened drawdowns; the protection mechanism amplifies the crisis.

> *YAAF v1.3 addresses the Liquidity Singularity with two systemic mechanisms: Correlation-Adjusted Scoring and Asynchronous Slash with dynamic thresholds.*

Chapter 2 — Theoretical Foundations

2.1 Kelly Criterion — Optimal Capital Growth

The Kelly Criterion (Kelly, 1956) is the foundational result in capital growth theory. Originally derived for information transmission by J.L. Kelly at Bell Labs, it was later applied to gambling and investment by Edward Thorp (Thorp, 1969) and formalized for financial markets in MacLean, Thorp & Ziemba (2011).

The Kelly fraction maximizes the expected logarithm of wealth — equivalently, it maximizes long-run geometric growth. For a binary bet with win probability W and win/loss ratio R:

```
f* = W - (1 - W) / R
```

YAAF uses the Kelly fraction as the theoretical ceiling for capital allocation, subject to adaptive scaling by track record length and confidence factor:

```
f_YAAF = f* × Kelly_scale(n)
```

$$\text{Kelly\_scale: } n<50 \rightarrow 0.25x \quad | \quad n<100 \rightarrow 0.33x \quad | \quad n \geq 200 \rightarrow 0.50x$$

The adaptive scaling responds to the well-documented observation (Thorp, 2008; MacLean et al., 2011) that Half Kelly is appropriate for established track records, while quarter or third Kelly is more conservative for short histories — reducing the risk of ruin during the track record building phase.

> *Academic grounding: Kelly (1956) proves that the Kelly fraction uniquely maximizes long-run geometric wealth growth. Breiman (1961) proves that any other strategy is asymptotically dominated by Kelly in the long run.*

2.2 Cramér-Lundberg Ruin Theory

The Cramér-Lundberg model (Cramér, 1930; Lundberg, 1903) provides a theoretical bound on the probability of ruin for a stochastic process with drift μ and volatility σ. For an initial capital u:

$$\psi(u) \approx \exp(-2\mu u / \sigma^2)$$

For the Markowitz Bot with μ=$2.41 per trade and σ=$55.61:

$$\psi(10{,}000) = \exp(-2 \times 2.41 \times 10{,}000 / 55.61^2) = 0.000017\%$$

This theoretical bound assumes normally distributed returns. With kurtosis=12.13 (fat-tailed), the assumption is violated and the bound is optimistic. YAAF therefore uses Bootstrap Monte Carlo as the primary validation method, with Cramér-Lundberg as a secondary theoretical reference.

> *Key insight: the theoretical ruin bound and the empirical bootstrap ruin rate both converge to 0.0% — not because fat tails are absent, but because Kelly-constrained position sizing (2.82% per trade) is small enough to absorb even extreme individual trade outcomes at the portfolio level.*

2.3 Deflated Sharpe Ratio — Anti-Overfitting Gate

The Sharpe Ratio (Sharpe, 1966) is the most widely used measure of risk-adjusted performance in quantitative finance. However, it has a critical vulnerability: multiple testing inflation. A researcher who evaluates 100 strategies will find approximately 5 with Sharpe > 1.96 by pure chance, even if all strategies have zero true edge.

López de Prado and Bailey (2014) derive the Deflated Sharpe Ratio (DSR), which corrects the Sharpe for non-normality of returns and the number of strategies tested:

$$\sigma\_SR_0 = \mathrm{sqrt}((1 - \mathrm{skew}\times SR + (\mathrm{kurt}-1)/4\times SR^2) / (T-1))$$

$$DSR = \Phi(SR / \sigma\_SR_0)$$

Where Φ is the cumulative normal distribution function, skew is the skewness of returns, kurt is the kurtosis, and T is the number of observations.

The DSR has three critical properties for YAAF:

- It corrects for fat tails — high kurtosis inflates σ_SR_0, making the gate more conservative for strategies with extreme return distributions.

- It corrects for multiple testing — with N=1 (single live system, no parameter search), DSR is at its most lenient; backtested systems with N>1 face a higher bar.
- It provides a probabilistic interpretation — DSR=99.9% means the probability that the observed Sharpe is genuine edge, not noise, is 99.9%.

> *For the Markowitz Bot: Sharpe=0.69, kurtosis=12.13, T=82 live trades, N=1. DSR = 99.999%. The fat tails inflated σ_SR_0 to 0.156, making the gate more conservative — yet the signal passed overwhelmingly. This is the correct response: penalize for fat tails, but recognize genuine edge.*

2.4 CVaR — Conditional Value at Risk

Value at Risk (VaR) measures the loss threshold exceeded with probability α. Conditional Value at Risk (CVaR), also known as Expected Shortfall (Rockafellar & Uryasev, 2000), measures the expected loss given that the loss exceeds VaR:

`CVaR_α = E[Loss | Loss > VaR_α]`

CVaR is preferred over VaR in YAAF for three reasons documented in the literature:

- Subadditivity — CVaR satisfies the subadditivity property required of coherent risk measures (Artzner et al., 1999), while VaR does not.
- Tail sensitivity — CVaR captures the severity of tail events, not just their frequency. For fat-tailed return distributions (kurtosis > 3), this distinction is critical.
- Incentive compatibility — penalizing CVaR discourages strategies that concentrate risk in rare catastrophic events while appearing safe by standard VaR measures.

YAAF applies a CVaR penalty in the S_RAW calculation, reducing the score for agents whose worst-case tail losses are disproportionate to their average trade size.

2.5 Markowitz Portfolio Theory and Correlation

Markowitz (1952) established that portfolio risk is not simply the sum of individual asset risks — correlation structure determines diversification benefits. A portfolio of assets with correlation ρ=1 provides no diversification; a portfolio with ρ=-1 can eliminate risk entirely.

YAAF v1.3 applies this insight to the agent pool: a vault deploying capital across N agents believes it has diversified, but if those agents share high pairwise return correlations, the effective number of independent strategies is much smaller than N.

The Pool Correlation Matrix (PCM) tracks capital-weighted pairwise correlations across all active agents, enabling the Correlation-Adjusted Score to penalize agents whose strategies converge with the existing pool.

Chapter 3 — S_RAW v4.1: The Scoring Engine

3.1 Design Philosophy

S_RAW is the statistical core of YAAF — a normalized score from 0 to 100 that captures the quality of an agent's trading performance across multiple dimensions. Each component is selected from the quantitative finance literature for its theoretical grounding and practical relevance to trading system evaluation.

The design follows three principles:

- Multi-dimensional — no single metric captures trading quality. A high Sharpe with extreme tail risk is different from a high Sharpe with controlled drawdown.
- Scale-free — metrics are normalized to remove dependency on account size, enabling comparison across agents with different capital levels.
- Anti-gamification — each component is paired with mechanisms that prevent artificial inflation through position sizing manipulation or cherry-picked periods.

3.2 The 10 Components

#	Component	Weight	Formula	Literature
1	Sharpe × DSR	18%	min(SR/2.5, 1.0)×100 × DSR_gate	Sharpe 1966; López de Prado 2014
2	Sortino × Confidence	8%	min(Sort/8.0, 1.0)×100 × min(n_loss/20, 1)	Sortino 1994
3	Win Rate	12%	WR × 100 (linear, no ceiling)	Van Tharp 1998
4	Profit Factor	15%	min((PF-1.0)/1.5, 1.0)×100	Schwager 1992
5	Avg R-Multiple	8%	min(AvgR/1.2, 1.0)×100	Van Tharp 1998
6	Expectancy E(R)	10%	min(ER/0.5, 1.0)×100 | ER=WR×AvgR-(1-WR)	Van Tharp 1998
7	Volatility	7%	max(1 - max_dd/100, 0)×100	Modigliani 1997
8	Stability²	12%	Consistency of monthly Sharpe, squared	Bailey & López de Prado 2014
9	Smoothness² (Calmar²)	12%	min(Calmar/3.0, 1.0)² × 100	Young 1991
DD Penalty	Drawdown Penalty	-25%	MaxDD% × 0.25 (direct subtraction)	Standard practice

3.3 Component Notes and Academic Justification

Sharpe × DSR (18%)

The Sharpe Ratio is the standard measure of risk-adjusted return, computed as the ratio of mean excess return to standard deviation. YAAF multiplies Sharpe by the DSR factor to penalize scores where the Sharpe may reflect statistical noise rather than genuine edge. An agent with DSR < 95% has its Sharpe contribution reduced proportionally.

The ceiling of Sharpe=2.5 is calibrated to professional hedge fund standards. A Sharpe above 2.5 is exceptional in live trading; allowing unbounded scores would disproportionately reward short track records with high but unstable Sharpe ratios.

Sortino × Confidence (8%)

The Sortino Ratio (Sortino & van der Meer, 1991) improves on Sharpe by penalizing only downside volatility. YAAF multiplies by a confidence factor based on the number of losing trades, as the Sortino ratio is unreliable when computed from fewer than 20 losing trades.

Expectancy E(R) — scale-free formulation (10%)

Expectancy in Van Tharp's formulation is E(R) = WR × AvgR - (1-WR), where AvgR is the average R-multiple (reward-to-risk ratio). Unlike dollar-based expectancy, this formulation is scale-free and does not depend on account size, enabling comparison across agents trading different instruments and sizes.

Stability² (12%)

Inspired by the consistency metric in Bailey & López de Prado (2014), Stability² measures the dispersion of monthly Sharpe ratios. An agent with consistent performance across months scores higher than one with similar average performance but high month-to-month variability. The squared exponent penalizes inconsistency more severely (following the principle that variance compounds in geometric growth).

Smoothness² — Calmar Ratio squared (12%)

The Calmar Ratio (Young, 1991) divides annualized return by maximum drawdown. YAAF uses it in squared form as a proxy for equity curve smoothness: an agent with Calmar=3.0 (excellent) scores 100 points; an agent with Calmar=1.5 scores 25 points ($0.5^2 \times 100$). The squared form rewards exponentially smoother equity curves.

3.4 Volume Axiom — Anti-Gamification

A critical vulnerability in any scoring system based on trade metrics is size manipulation: an agent can trade tiny positions to generate a high win rate and Sharpe ratio with minimal real risk, then use the resulting score to access vault capital far beyond what the track record justifies.

The Volume Axiom addresses this by requiring a minimum average position size proportional to the agent's stake:

```
Volume_floor = Initial_Balance × 0.001
Volume_OK = avg_abs_profit >= Volume_floor
```

An agent with a FLAGGED Volume Axiom status cannot access vault capital regardless of its S_RAW score. This ensures that all scores are backed by meaningful risk-taking, not paper trading at microscopic scale.

3.5 S_RAW Formula

> **S_RAW = Σ(component_i × weight_i) - DD_Penalty**
>
> **S_RAW = (Sharpe_DSR × 0.18 + Sortino_conf × 0.08 + WR × 0.12 + PF × 0.15**
>
> **+ AvgR × 0.08 + ER × 0.10 + Vol × 0.07 + Stab² × 0.12 + Smooth² × 0.12)**
>
> **- MaxDD% × 0.25**

S_RAW is bounded to [0, 100]. An agent with perfect scores across all positive components and zero drawdown achieves S_RAW=100. A real-world agent with excellent performance typically scores 60-90; 40-60 is good; below 20 indicates a difficult period or insufficient track record.

Chapter 4 — SISTEMA Score and Capital Allocation

4.1 From S_RAW to SISTEMA

S_RAW captures the quality of a single scoring period — a snapshot of performance. The SISTEMA score (0-1000) accumulates performance over time, incorporating the agent's historical trajectory with memory of past performance and penalties for inconsistency.

> **SISTEMA = EMA × CF × SF**
>
> **EMA(t) = EMA(t-1) × 0.85 + S_RAW × 10 × 0.15**
>
> **CF = min(sqrt(total_trades / 200), 1.0)**
>
> **SF = min(worst_S_RAW_6_batches / 40, 1.0)**

The three components serve distinct functions:

- EMA — Exponential Moving Average with α=0.15 provides memory of past performance (85% weight on history) while incorporating new information. Initial EMA=300 prevents cold-start bias.
- CF (Confidence Factor) — scales the score by the square root of trade count, reaching CF=1.0 at 200 trades. This reflects the statistical principle that confidence in a Sharpe estimate scales with the square root of the sample size.
- SF (Safety Factor) — penalizes agents whose worst batch in the last 6 scoring rounds scored below 40. This captures regime degradation and strategy deterioration that might not be visible in the cumulative EMA.

4.2 Stage Classification

Stage	SISTEMA Range	Capital Multiplier	Min Stake	Max Slash β
EXPERIMENTAL	0 – 199	0.5×	50 YLD	100%
PROMISING	200 – 399	2.0×	200 YLD	100%
VERIFIED	400 – 599	5.0×	500 YLD	75%
ELITE	600 – 799	10.0×	1,000 YLD	50%
LEGENDARY	800 – 1000	20.0×	2,000 YLD	50%

4.3 Capital Allocation Formula

Capital allocation is determined by the minimum of two independent constraints: the Kelly-derived maximum and the stage-based maximum. Both constraints must be satisfied simultaneously.

```
C_allocated = min(C_kelly, C_stage)
C_kelly = f* × Kelly_scale(n) × CF × V_vault
C_stage = M(stage) × S_agent
```

This dual-constraint design ensures that:

- An agent cannot access capital beyond what its demonstrated edge justifies (Kelly constraint).
- An agent cannot access capital beyond what its current stage and stake commitment support (Stage constraint).
- The only path to higher capital allocation is genuine performance improvement over time.

4.4 Minimum Stake Formula

Stake requirements are derived from the agent's performance metrics, not set arbitrarily. The minimum stake scales with the requested capital allocation and the agent's historical drawdown:

```
S_min = max(S_floor(stage), C_request × MDD_factor × safety(stage))
MDD_factor = 0.40 - 0.25 × (S_RAW / 100)
```

A higher S_RAW reduces the MDD_factor, lowering the stake requirement — better-performing agents need less collateral per unit of capital. Safety multipliers by stage (EXPERIMENTAL=2.0, LEGENDARY=1.0) reflect the higher uncertainty in early-stage track records.

Chapter 5 — The Slash Mechanism

5.1 Design Principles

The slash mechanism is YAAF's enforcement layer — the economic consequence for agents that underperform or violate the accountability framework. Effective slash design must balance three competing requirements:

- Protection — the vault must be protected from significant capital loss by agent failure.
- Proportionality — slash penalties must be proportional to the harm caused, not arbitrary.
- Gaming resistance — the mechanism must not create incentives for strategic behavior that harms the vault while preserving agent stake.

5.2 Dual-Trigger Slash Function

YAAF v1.2 introduced the dual-trigger slash that fires on either drawdown OR capital loss, whichever produces the higher slash amount:

> **Slash = max(Slash_DD(δ), Slash_CapLoss(loss / C_alloc))**
>
> **Slash_DD: 0 if δ<10% | 0.4×(δ-0.10)×S if 10-25% | β×S if δ≥25%**
>
> **Slash_CapLoss: 0 if loss<10% | 0.4×(loss%-0.10)×S | β×S if loss≥20%**

The dual trigger closes the 24% loophole: an agent that strategically stops trading at 24% DD to avoid the 25% threshold is still slashed on the capital loss dimension if it has already destroyed 20% of allocated capital.

Drawdown	Capital Loss	Slash (50 YLD stake)	Trigger Mechanism
5%	10%	0 YLD	Both below threshold — safe zone
5%	20%	10 YLD	Capital loss trigger fires
15%	10%	1.0 YLD	DD trigger: 0.4×0.05×50
24%	10%	2.8 YLD	DD trigger — agent stops early
24%	20%	10 YLD	Capital loss overrides — gap closed
25%	any	50 YLD	Full slash — critical threshold

5.3 Slash Distribution — The Social Contract

YAAF establishes a three-way distribution of slashed stake that aligns incentives across all protocol participants:

> **50% burned (YLD deflationary mechanism)**
>
> **30% to vault depositors (loss compensation)**

> **20% to UBI pool (verified humans)**

The UBI component is not an afterthought. When autonomous agents manage capital at scale and fail, a portion of the economic consequence of that failure flows to the humans the protocol serves. This is the social contract at the heart of the Yelden Protocol: machines pay for their mistakes, and humans benefit from that accountability.

> *The UBI component is also economically rational: it creates a permanent incentive for verified humans to participate in the protocol as accountability beneficiaries, providing a growing pool of stakeholders with direct financial interest in maintaining protocol integrity.*

5.4 Dynamic Slash Threshold — YAAF v1.3

YAAF v1.3 extends the slash mechanism with a dynamic threshold that scales inversely with systemic correlation:

> **δ_slash(t) = δ_base + β × SCI(t)**
>
> **SCI(t) = capital-weighted average pairwise correlation across pool**

When SCI is high (agents are highly correlated), the slash threshold expands — providing more buffer before slash fires. This prevents the cascade scenario where simultaneous drawdowns across correlated agents trigger simultaneous slash events, amplifying the systemic crisis.

SCI (pool stress)	δ_slash effective	vs baseline	Interpretation
0.00 (decorrelated)	25.0%	unchanged	Normal conditions
0.20 (low stress)	28.0%	+3.0pp	Minor co-movement
0.50 (elevated)	32.5%	+7.5pp	Correlated market
0.80 (high stress)	37.0%	+12.0pp	Crisis conditions
1.00 (convergence)	40.0%	+15.0pp	Maximum protection

Chapter 6 — Systemic Risk: The Liquidity Singularity Problem

6.1 From Individual to Systemic Accountability

YAAF v1.2 established individual accountability: each agent's stake, capital allocation, and slash conditions are derived from its own trading record. This is necessary but not sufficient for a protocol managing capital across multiple agents simultaneously.

The Liquidity Singularity problem — the convergence of correlated agents into simultaneous positions — requires systemic accountability: infrastructure that measures and penalizes the contribution of individual agents to pool-level risk.

6.2 Mechanism 1 — Correlation-Adjusted Score

The Correlation-Adjusted Score extends S_RAW with a diversification factor that rewards genuine strategy differentiation:

$$S\_final = S\_RAW \times D\_factor(\rho)$$

$$D\_factor(\rho) = 1 - \alpha \times \max(0, \rho - \rho\_threshold)$$

Parameters: ρ_threshold=0.50 (no penalty below moderate correlation), α=0.80 (penalty intensity). A fully correlated agent (ρ=1.0) loses 40% of its score — making diversification a first-class economic incentive.

The correlation ρ is computed from the Pool Correlation Matrix (PCM), updated every epoch (default 7 days). Each agent submits normalized period returns; the PCM is maintained by the registry contract.

6.3 Mechanism 2 — Asynchronous Slash

The Systemic Correlation Index (SCI) aggregates the PCM into a single measure of pool-level stress:

$$SCI(t) = (2/N(N-1)) \times \Sigma\_\{i<j\}\ w\_ij \times \rho\_ij(t)$$

The capital-weighted SCI drives the dynamic slash threshold described in Chapter 5. One data structure — the PCM — powers both mechanisms, minimizing oracle surface area and trust assumptions.

6.4 Academic Grounding for Systemic Risk Mechanisms

The Correlation-Adjusted Score draws on the risk decomposition literature (Qian, 2006; Maillard et al., 2010) which demonstrates that equal-weight portfolios of correlated assets dramatically underperform in terms of diversification efficiency. The penalty for correlation convergence is directly analogous to the risk contribution penalty in Risk Parity portfolio construction.

The dynamic slash threshold draws on the literature on procyclicality in financial regulation (Adrian & Brunnermeier, 2016; Brunnermeier & Pedersen, 2009). Fixed margin requirements amplify crises by forcing deleveraging precisely when markets are most stressed. YAAF v1.3 inverts this dynamic: thresholds expand under stress, reducing the procyclical pressure of simultaneous slashing.

> *"The failure of risk management in 2008 was not primarily a failure of models. It was a failure of systemic thinking — the inability to recognize that individually sound risk management decisions become collectively destructive when everyone makes them simultaneously." — Brunnermeier & Pedersen (2009), adapted*

Chapter 7 — Empirical Validation: The Markowitz Bot

7.1 Why Empirical Validation Matters

Mathematical frameworks for trading systems are trivially easy to construct and catastrophically easy to overfit. The literature is replete with strategies that perform beautifully in simulation and fail in live trading. YAAF is validated against live trading data — not simulations, not backtests, not paper trading.

The Markowitz Bot (ERC-8004 Agent #27703, Polygon Mainnet) is the reference implementation of YAAF. All parameters are calibrated to its production data.

7.2 Live Trading Statistics

Metric	Value	Significance
Trades (live)	122	Short track record — honest CF=0.78
Win Rate	58.68%	Positive edge confirmed
Profit Factor	1.03 (stress period)	Near breakeven — difficult week
Sharpe Ratio	0.09 (stress) / 1.44 (normal)	Honest — stress periods visible
DSR (N=1 live)	81.67% / 99.999%	Varies by period, N=1 throughout
Max Drawdown	3.47% – 6.81%	Well inside 25% threshold
Skewness	-0.64	Slight left skew — modest tail risk
Kurtosis	12.13	Fat tails — bootstrap MC required
Kelly f*	0.00 – 0.49	Varies by period performance
Ruin Rate (bootstrap)	0.0%	Confirmed empirically — 5,000 paths

7.3 Bootstrap Monte Carlo — Why Gaussian Simulation Fails

Standard Monte Carlo simulation assumes normally distributed returns. For the Markowitz Bot with kurtosis=12.13 (four times above normal), this assumption is violated. A Gaussian simulation would dramatically underestimate ruin risk.

Bootstrap Monte Carlo resampling draws directly from the 122 real trades, preserving the actual fat-tail distribution. The procedure:

- Sample 200 trades with replacement from the actual trade history.
- Simulate portfolio evolution under Kelly-constrained position sizing.
- Record whether the portfolio reaches ruin (defined as 100% capital loss).
- Repeat 5,000 times to build the empirical ruin distribution.

Result: 0.0% ruin rate across all 5,000 paths, across all SISTEMA stages from EXPERIMENTAL to LEGENDARY.

> *The kurtosis insight: kurtosis=12.13 implies fat tails that would normally inflate ruin risk dramatically. But Kelly-constrained position sizing (2.82% per trade) is small enough that*

> *even the worst-case trade of -$142 represents only 0.04% of a $10,000 account at the EXPERIMENTAL allocation of $25. Fat tails exist at the trade level; they are absorbed by the Kelly constraint at the portfolio level.*

Stage	Bootstrap Ruin%	Gaussian Ruin%	Bootstrap CVaR95	C_alloc	Kelly Scale
EXPERIMENTAL	0.0%	0.0%	$9,999.94	$25	0.33×
PROMISING	0.0%	0.0%	$9,999.03	$400	0.33×
VERIFIED	0.0%	0.0%	$9,995.47	$1,863	0.33×
ELITE	0.0%	0.0%	$9,995.47	$1,863	0.33×
LEGENDARY	0.0%	0.0%	$9,995.47	$1,863	0.33×

Chapter 8 — Protocol Architecture

8.1 System Layers

Layer	Function	Status
Oracle	S_RAW v4.1 + DSR scoring for any MT5 agent. Open-source, verifiable, broker-agnostic.	LIVE
Identity	ERC-8004 on-chain agent identity. Persistent accountability across deployments.	LIVE (#27703)
Contracts	YLDToken + AIAgentRegistry on Polygon. Score submissions on-chain.	LIVE (Mainnet)
Dashboard	yelden.fund/agent — real-time agent data from VPS.	LIVE
Join Portal	join.yelden.fund — any MT5 agent can submit for YAAF scoring. Leaderboard + Diagnosis.	LIVE
Staking	YAAF stake → capital multiplier. Slash on DD/capital loss violations.	Framework ready
Vault	$10,000 initial capital allocated by score to top agents.	Q3 2026
UBI	20% of slashed YLD to verified humans. Machines pay for mistakes.	Q3 2026
Multi-scorer	Genesis Scorer Program — 5-10 independent scorers for consensus.	Q2 2026

8.2 Smart Contracts — Polygon Mainnet

YLDToken: 0xE5Cc95489eC4Afb957f5C71593f6e4e028410cd4

AIAgentRegistry: 0x32F534265090d8645652b76754B07E6648b51571

Markowitz Bot: 0xfD3d7fdda54360Dc29CAa2f746aD77278A266cFc

8.3 The Genesis Scorer Program

In Phase 1 (Genesis), the protocol operates with a single centralized scorer. This is an intentional design choice for bootstrap — the system requires a reference scorer before decentralizing.

Phase 2 (Multi-Scorer Consensus, Q2 2026) distributes scoring across 5-10 independent operators. Score accepted when ≥2/3 of active scorers agree within ±5 S_RAW points. Scorers with uptime <99% lose compensation for that round.

> *The centralized scorer is the protocol's acknowledged primary vulnerability in the Genesis phase. The Genesis Scorer Program opening in May 2026 is the critical path to addressing this vulnerability.*

Chapter 9 — Comparison with Existing Frameworks

Feature	Eth PoS	Cosmos	EigenLayer	AgentProof	YAAF v1.2	YAAF v1.3
Slash basis	Consensus	Consensus	Service SLA	Reputation	P&L + CapLoss	P&L + CapLoss
Capital formula	Fixed 32 ETH	Fixed	Variable	None	Kelly × CF	Kelly × CF
Ruin bound	None	None	None	None	C-L + Bootstrap	C-L + Bootstrap
Tail risk	None	None	None	None	CVaR+Skew +Kurt	CVaR+Skew +Kurt
Anti-overfitting	N/A	N/A	N/A	None	DSR gate	DSR gate
Anti-gamification	None	None	None	None	Volume Axiom	Volume Axiom
Corr. penalty	No	No	No	No	No	D_factor(ρ)
Systemic risk index	No	No	No	No	No	SCI(t)
Dynamic slash	No	No	No	No	No	δ_base+β×SCI
UBI component	No	No	No	No	20% of slash	20% of slash
Live validation	Yes	Yes	Partial	No	122 trades, MC	122 trades, MC

Chapter 10 — Acknowledged Limitations and Open Questions

10.1 Short Track Record

122 live trades is a short track record by institutional standards. The DSR with N=1 (single live system) is generous — a strategy with 10 parameter variations would face a much higher DSR bar. The Confidence Factor (CF=0.78 at 122 trades) explicitly penalizes the SISTEMA score for this uncertainty.

The honest path: continue operating, accumulate trades, let the framework evaluate the track record as it grows.

10.2 Single Centralized Scorer

The current architecture depends on a single scorer operated by the protocol founder. This is the primary trust vulnerability. The Genesis Scorer Program (May 2026) is the critical path to resolution.

10.3 Cramér-Lundberg Assumption Violation

The theoretical ruin bound assumes normally distributed returns. With kurtosis=12.13, this assumption is violated and the bound is optimistic. Bootstrap Monte Carlo is the honest validation method — and both methods converge to 0.0% ruin, though for different reasons.

10.4 Large Vault Slash Mechanism

The slash function for vaults with C_alloc > $100,000 is under active research. At large scale, even a partial slash may create market impact that affects YLD token price, which feeds back into collateral values — a non-linear feedback not captured in current formulas.

10.5 Oracle Integrity for Correlation Matrix

The PCM depends on agents submitting honest period returns. Agents with incentive to appear decorrelated could submit manipulated returns. Mitigation: returns cross-validated against on-chain P&L. Full solution requires trusted execution environment or zero-knowledge proofs of return correctness — deferred to v1.4.

10.6 Safety Factor (SF) Implementation

The SF requires historical batch scores to penalize agents with recently deteriorating performance. Current implementation is functional for agents with multi-period history; newly onboarded agents receive SF=1.0 by default until sufficient history accumulates.

Conclusion

YAAF v1.3 is the product of four rounds of peer review, empirical validation against 122 live trades, and systematic engagement with the foundational literature in mathematical finance and risk management.

The framework is simultaneously:

- Optimal — Kelly criterion maximizes long-run geometric wealth growth.
- Accountable — stake commitment proportional to risk taken and damage caused.
- Honest — DSR validates that scores reflect genuine skill, not statistical noise.
- Tail-aware — CVaR, skewness, and kurtosis penalties for fat-tail strategies.
- Anti-gamification — Volume Axiom prevents score inflation via tiny positions.
- Complete — capital loss trigger closes the 24% DD loophole.
- Systemic — Correlation-Adjusted Score and Asynchronous Slash address the Liquidity Singularity.
- Empirically grounded — all parameters validated against live trading data.

The deeper insight is this: the problem of autonomous agent accountability is not fundamentally different from the problem of human fund manager accountability. The tools exist — Kelly, DSR, CVaR, Monte Carlo. What has been missing is their systematic application to the autonomous agent context, implemented on-chain with economic consequences.

YAAF does not make autonomous agents trustworthy by assertion. It makes them accountable by mathematics.

> *The UBI component is not an afterthought. When AI agents manage capital at scale, a portion of the proceeds of their failures should flow to the humans they serve. That is the social contract at the heart of the Yelden Protocol.*

Yelden Protocol — yelden.fund

March 2026 | Polygon Mainnet | ERC-8004 Agent #27703

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